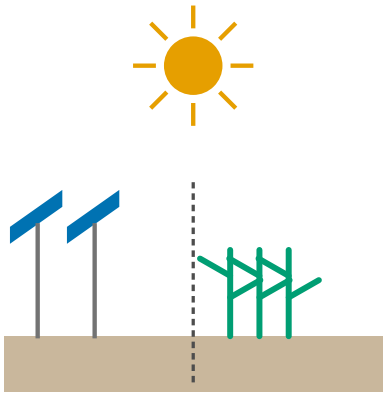
❶ How much solar fits on India's farmland and keeps 80% of the harvest?

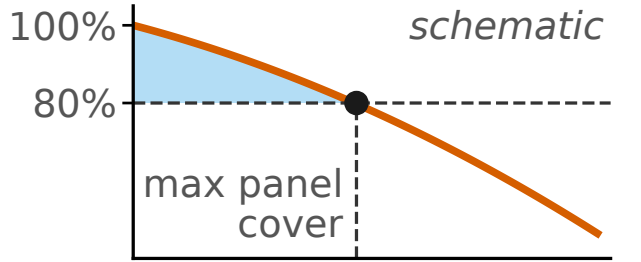


❷ Simulate shade → harvest

shade model → crop model

harvest (% of open field)

schematic

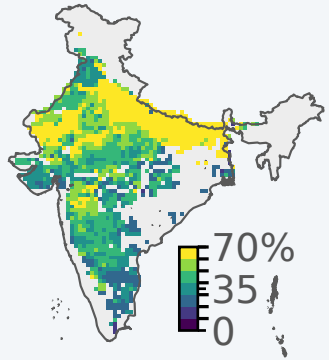


panel cover: sparse → dense

every 0.25° cell · 5 climate models

❸ Answer for India

max panel cover keeping 80% harvest



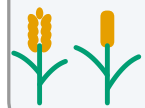
45.5 TWp

would generate ≈ 48× India's yearly electricity use

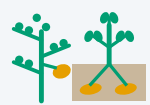
(upper bound)

2050 (SSP5-8.5): −1% vs 1995–2014

max panel cover:



~61% cereals



~29% legumes